9.1.2 ERC20 token standard

As an immediate response to introduction of Ethereum, numerous cryptocurrency tokens emerged to represent various services and businesses. The ERC20 standard interface was introduced so that Ethereum-based cryptocurrency tokens follow a standard and are compatible. The ERC20 standard specifies a set of rules that allows the tokens to interact with one another, exchange with one another, and transact on the Ethereum network.

The OpenZeppelin organization (https://openzeppelin.com) is an active community that supports Ethereum protocol. Its improvements and token standards are discussed at https://docs.openzeppelin.com/contracts/3.x/api/token/erc20#IERC20. Here is a partial definition of the ERC20 interface:

```
contract ERC20 {    function totalSupply() public view returns (uint256);    function
balanceOf(address tokenOwner) public view returns (uint256
                                        balance);    function allowance(address
tokenOwner, address spender) public view
                           returns (uint256 remaining);    function transfer(address to,
uint256 tokens) public returns
                                                              (bool success);    function
approve(address spender, uint256 tokens) public returns
                                                              (bool success);    function
transferFrom(address from, address to, uint256 tokens)
                                                 public returns (bool success);    //events
event Transfer(address indexed from, address indexed to,
                                                             uint256 tokens);    event
Approval(address indexed tokenOwner, address indexed spender,
                                                             uint256 tokens);  }
```

The ERC20 definition also includes the token `name`, token `symbol`, and an attribute (`decimal`) that specifies how to represent a fractional of a token--the scale factor. To create and deploy a token for an asset or utility compliant with the ERC20 standard, you'll implement a smart contract with the functions required by the ERC interface: